

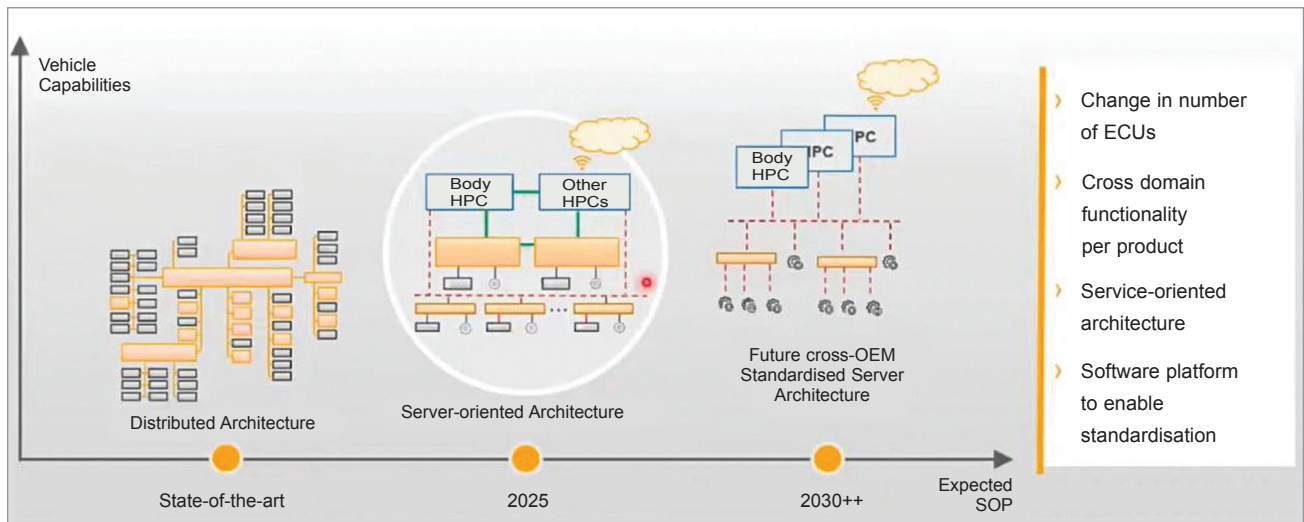


Fig. 4: Vehicle capabilities increasing with time

needed from the vehicle architecture as of now. With this computational revolution, we are talking about a complete shift in the way we will implement and design vehicle architecture.

Talking about today's vehicles, every ECU works for itself and shares only necessary information with other ECUs. Moreover, it is a closed system, which means one cannot upgrade the software and other aspects. Going forward, we are going to witness a big change where a central ECU will be able to decouple software from hardware.

Today, software is very much attached to the hardware such that the software cannot function in another hardware properly, or if there is any change in the hardware. This makes it inconvenient to upgrade hardware as one also needs to work closely on the software. We need to decouple hardware and software so that changes/modifications can be made in either of them independently.

With high computational complexity, there is a high chance we might bring in cross-domains like human-machine interface (HMI) and infotainment systems, which are basically different domains but will have to come together.

Further, central computation will be enabled in near future, which would reduce the number of ECUs. This, in turn, would reduce complexity, the interfaces, and the cost. And it would allow us to easily implement

plug-and-play type of product development with decoupled hardware and software.

We can see this trend of zonal ECUs and central ECUs already, and many OEMs have already migrated to this kind of architecture. Talking about the time to market, we are still looking at about the same time or in fact reduced time.

Change in role of ECUs

With improved architecture, as discussed, the number of ECUs will come down and upgradability will improve. It is difficult to update an ECU and predict its impact on the ECUs at present.

With increased computational power, we will be able to introduce new features like cross-domain functions. The software platform will become standard and with that the merging product line will be supported with service oriented architecture. Then the location of function in the vehicle will no longer matter, as long as it is within the vehicle or is a part of the offboard function.

With all this, what are the things we will be able to address?

We will be able to update firmware over the air, so updating multiple ECUs will become easier. The associated data will come in the central ECU, which will help us in predictive diagnosis. Apart from this, there will be an increasing demand for virtualisation.

It is important that we have this support in the central ECU, so that

we can support JAVA, Android, etc based applications in future with the available hardware. We have to make sure that the hardware is capable of doing such things and the software framework allows us to do it.

The challenge is to bring all this together on a state-of-the-art integration platform because, ultimately, the integration becomes a key factor for OEMs. They would like to have faster development as there will be multiple parties involved, who would have to be brought together with a robust integration platform.

There will be a central ECU acting as a brain or central processing unit, and there will be zonal ECUs for their respective zones. If we take the example of a window lifter, when there is rain the system would have to be smart enough to close the sun-roof, raise windows, and switch on the wiper without any user intervention.

There are two ways to achieve this kind of operation. Either the zonal controllers along with the smart sensors and actuators could do it, or the high-performance central ECU could decide and send signals, so that the zone controllers do their part. Based on various considerations, one can enable any of these two architectures.

The way forward

The current situation is like this: We have multiple ECUs that are connected but are doing their own part independently, exchanging information only when needed. Going forward, these